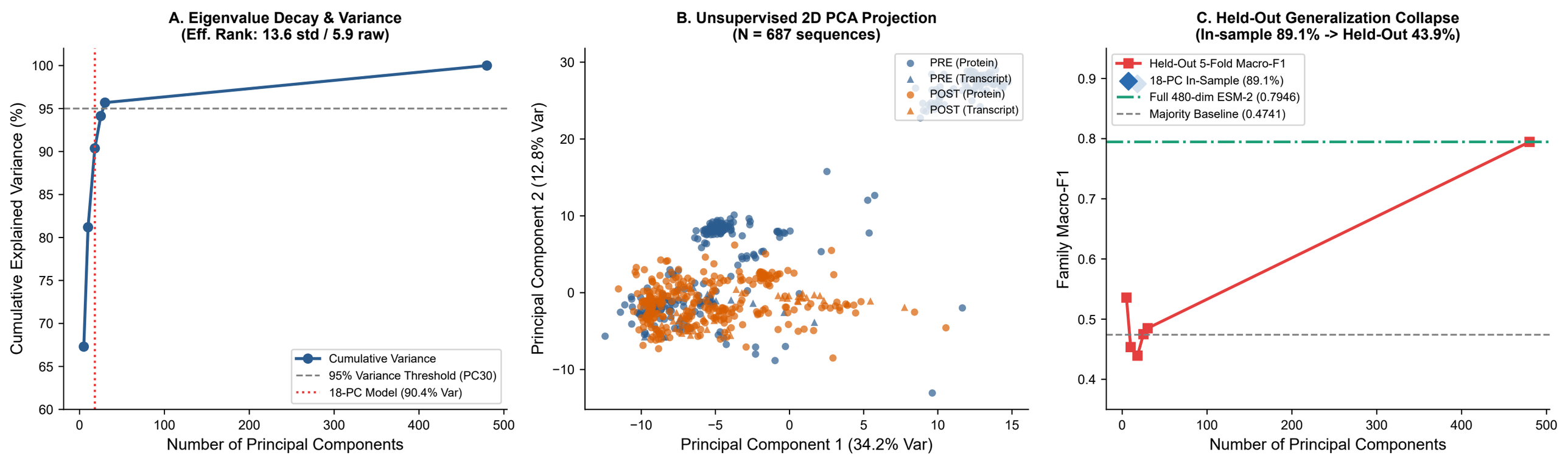

A. Eigenvalue Decay & Variance
(Eff. Rank: 13.6 std / 5.9 raw)
Cumulative Explained Variance (%)
Number of Principal Components
Cumulative Variance
95% Variance Threshold (PC30)
18-PC Model (90.4% Var)
B. Unsupervised 2D PCA Projection
(N = 687 sequences)
Principal Component 2 (12.8% Var)
Principal Component 1 (34.2% Var)
PRE (Protein)
PRE (Transcript)
POST (Protein)
POST (Transcript)
C. Held-Out Generalization Collapse
(In-sample 89.1% -> Held-Out 43.9%)
Family Macro-F1
Number of Principal Components
Held-Out 5-Fold Macro-F1
18-PC In-Sample (89.1%)
Full 480-dim ESM-2 (0.7946)
Majority Baseline (0.4741)